

MS based Proteomics

Techno Jargon and when it comes from



Different programs will sometimes define these differently.

- This is supremely frustrating and confusing!!



Let's look at Techno Jargon . Some specific to Spectronaut

PG= Protein Group

Pep = Peptide

EG = Elution Group

FG = Fragment Group (this is just for isobaric or spike in experiments)

PTM = Post translational Modification

XIC = Extracted Ion Chromatogram

AA = Amino Acid

RT = Retention Time

IM = Ion Mobility

Precursor (more on this in a bit)

DIA – Data independent Acquisition.

M or m = mass

Z = charge (# of protons)

m/z = mass divided by charge

H+H = mass with the proton (confusing as we don't actually have a hydrogen)

MS1 = intact mass of the peptide ion

MS2 = fragment mass broken off from the peptide ion



Lets look at what the mass spectrometer really measures

- We measure ions
- For proteomics these are usually peptide ions with 1-3 protons attached



People are still trying to figure out the how making ions actually this happens!



Formation of Gaseous Peptide Ions from Electrospray Droplets: Competition between the Ion Evaporation Mechanism and Charged Residue Mechanism

Elnaz Aliyari and Lars Konermann*

Cite this: *Anal. Chem.* 2022, 94, 21, 7713–7721

Publication Date: May 19, 2022

<https://doi.org/10.1021/acs.analchem.2c01355>

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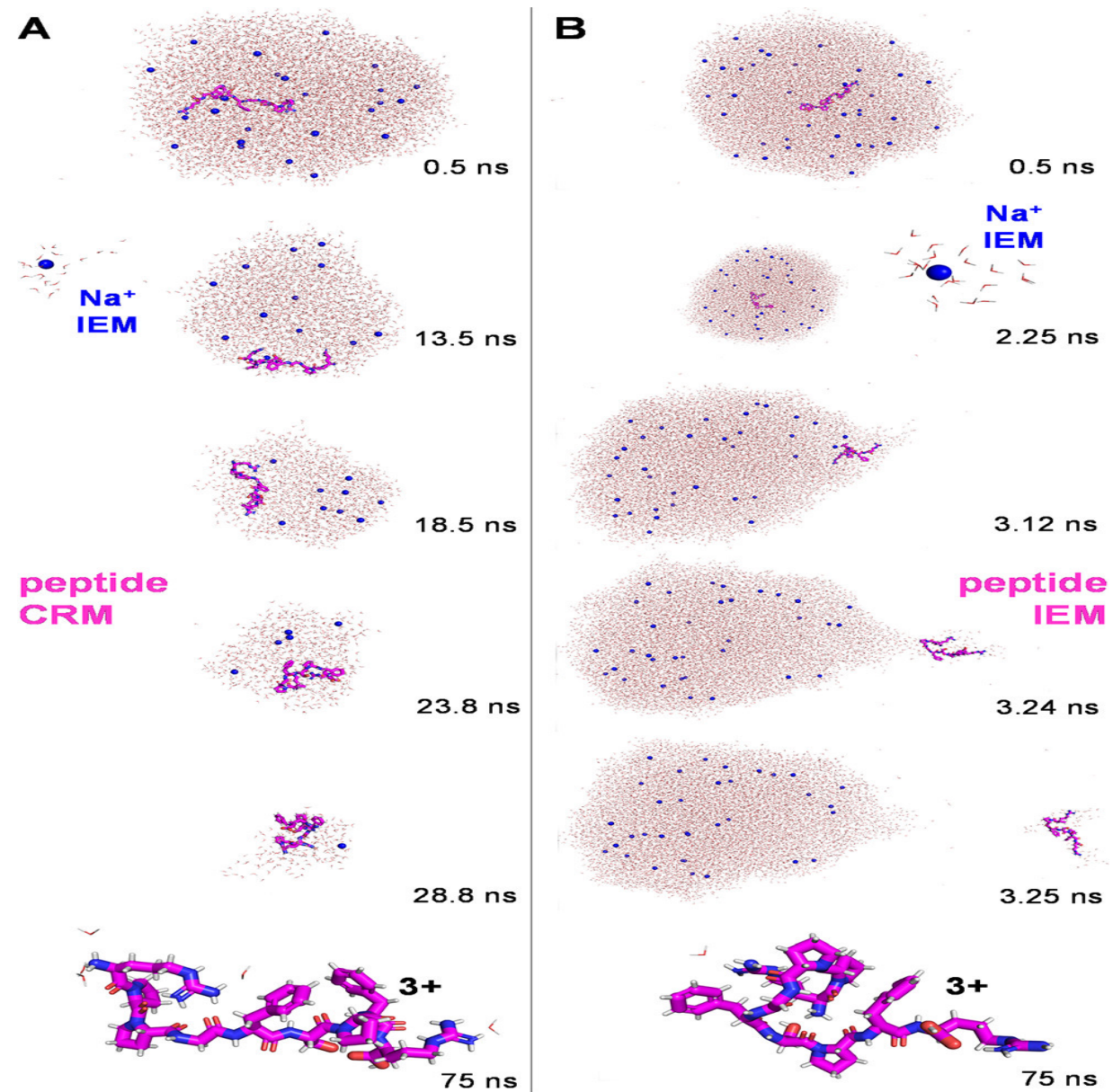
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<https://pubs.acs.org/doi/full/10.1021/acs.analchem.2c01355>





Anyway...

After this you get peptide with protons on them
in the gas phase = ions.

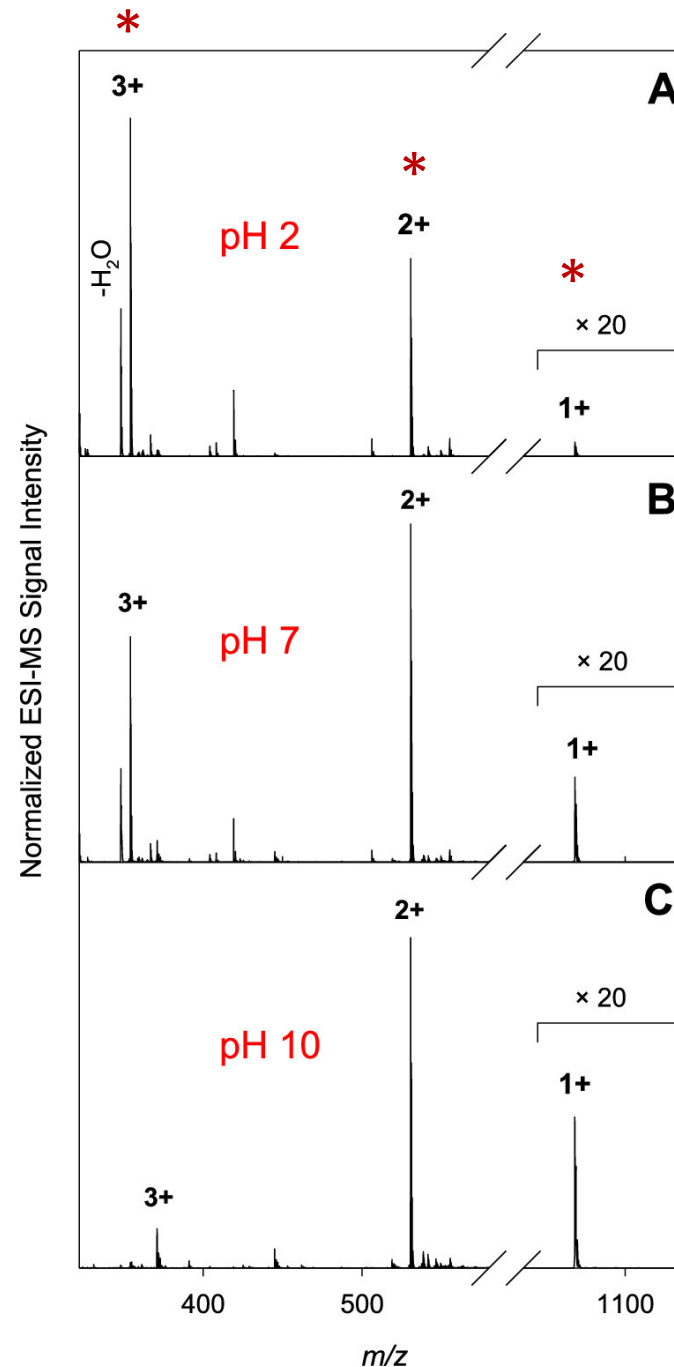
This is what we try and measure!!

!!And mass spectrometers do not measure
mass! They measure mass over charge.
We cannot measure neutrals!!



Most software calls
these precursors as
the same peptide
can exist in multiple
charge states

Precursors will have
a different “mass” as
we actually measure
 m/z (mass / charge)



*

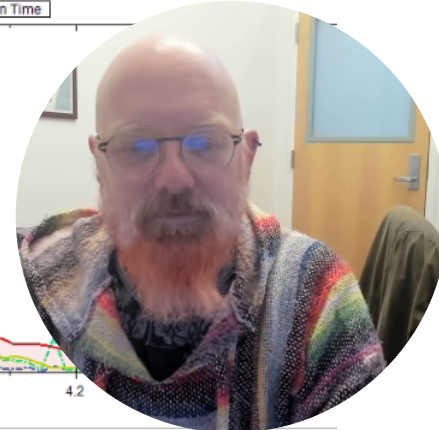
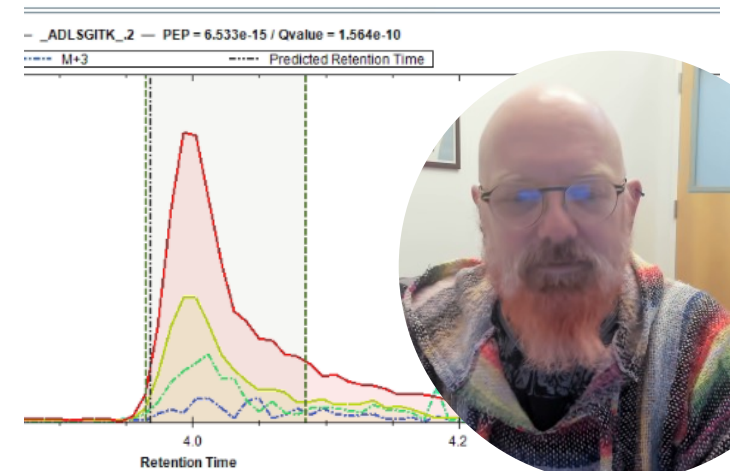
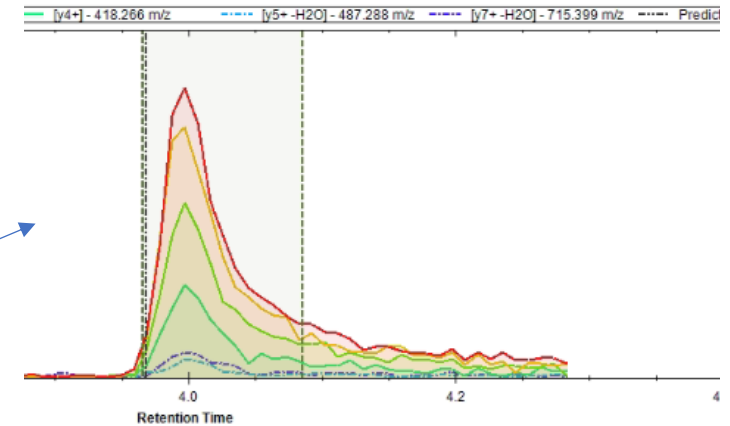
The LC part

Peptides come out of a machine that separates peptides in time usually by their hydrophobicity using liquid chromatography

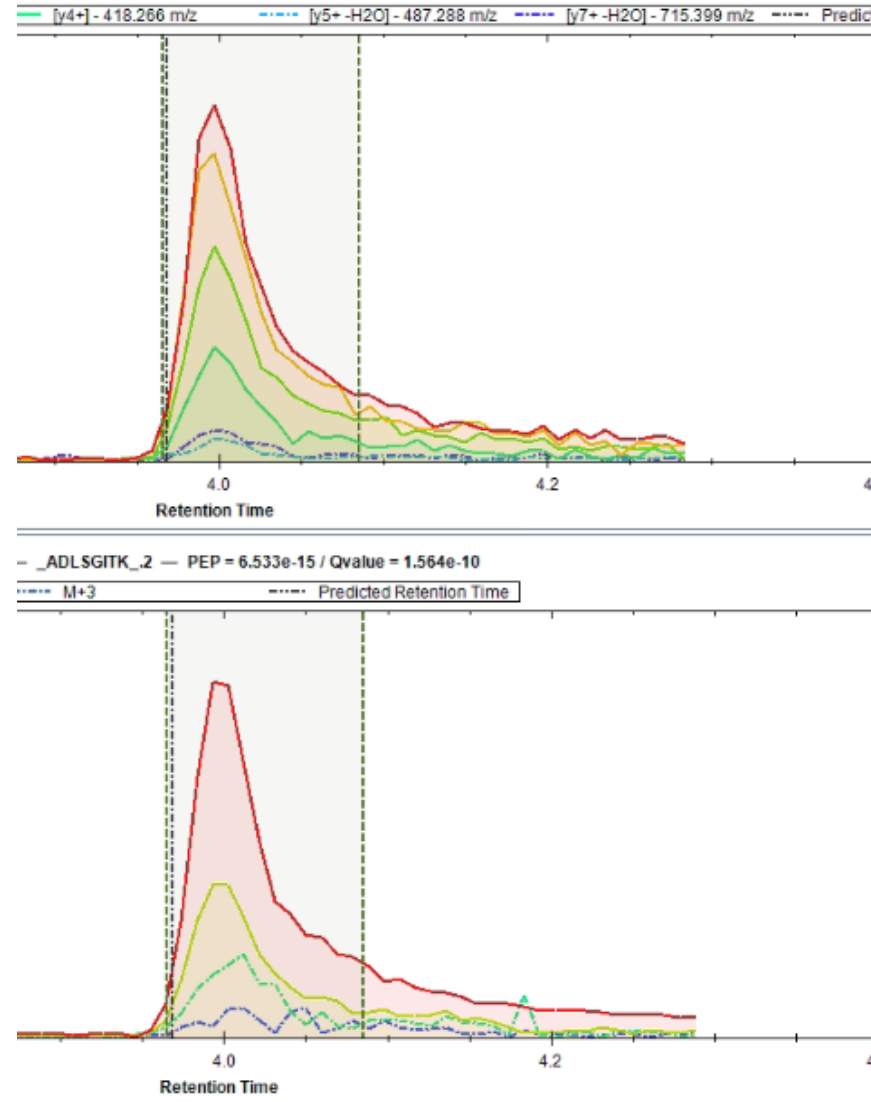
This is called HPLC or UPLC

This makes
what we call
peaks!

Think of them
as analytes



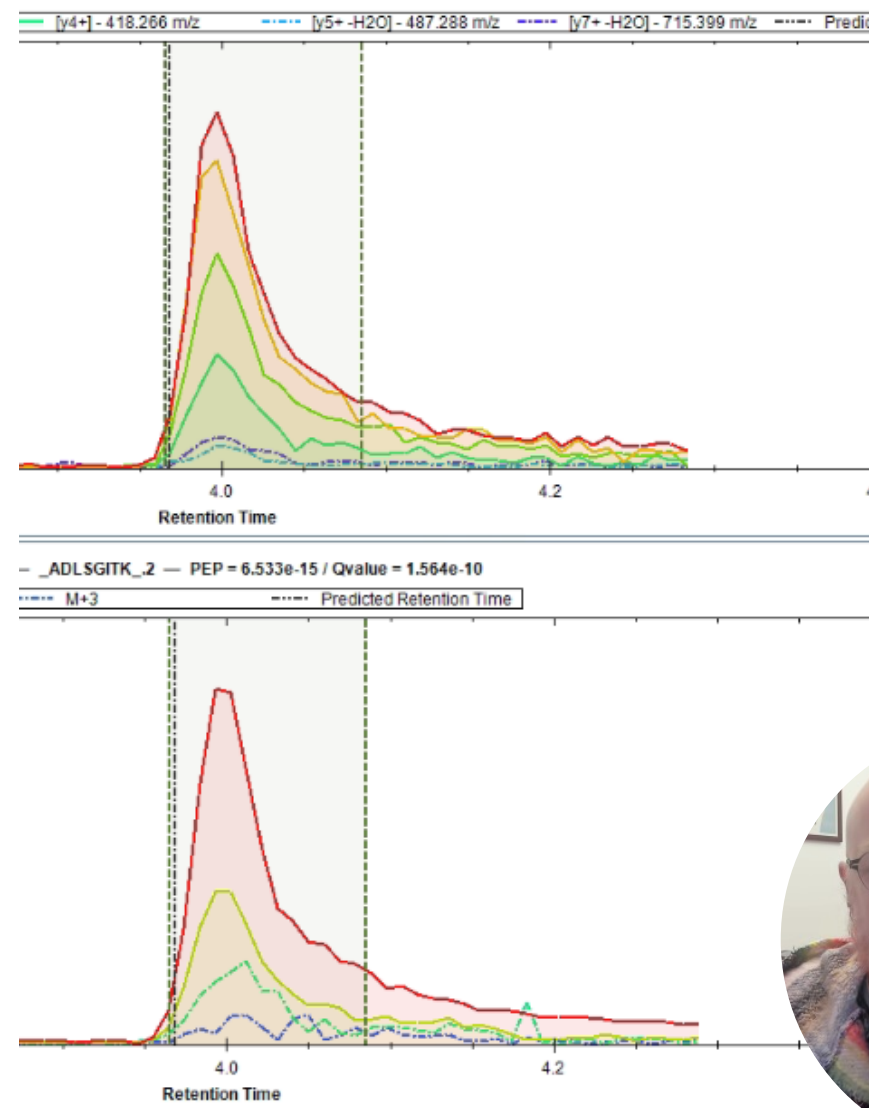
We measure either the area or the height of these peaks



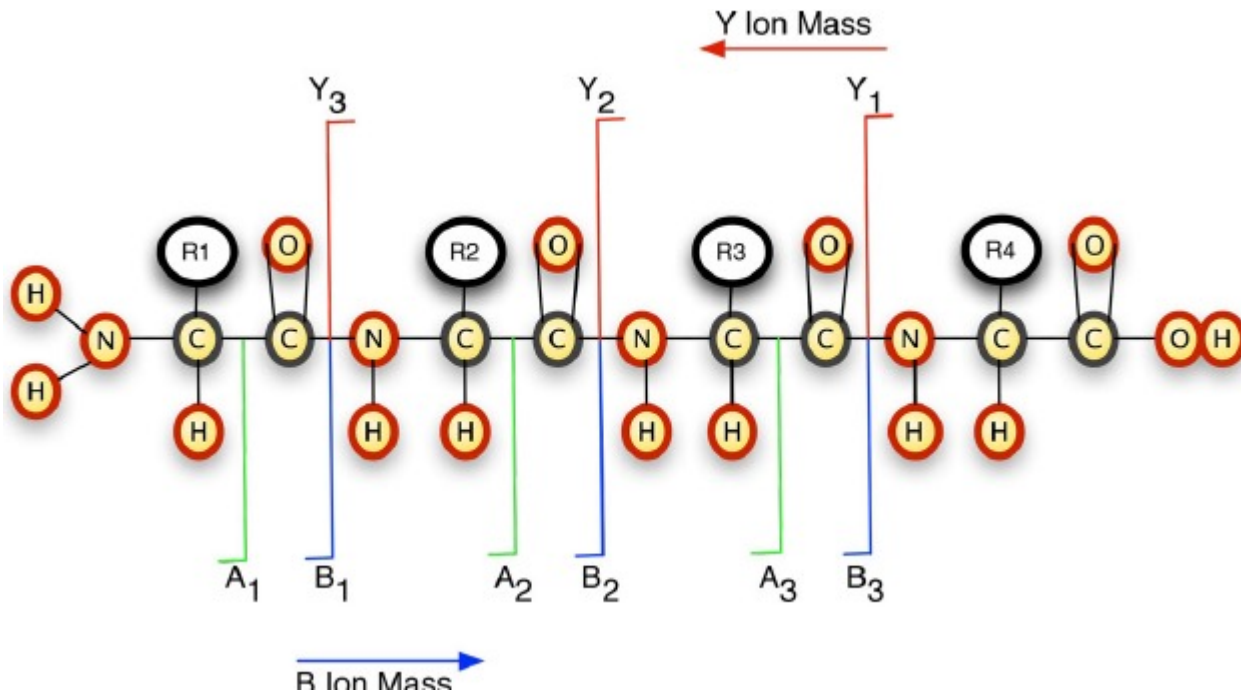
Each peak has lots! of Ions we can measure

And we Measure all of them!!!

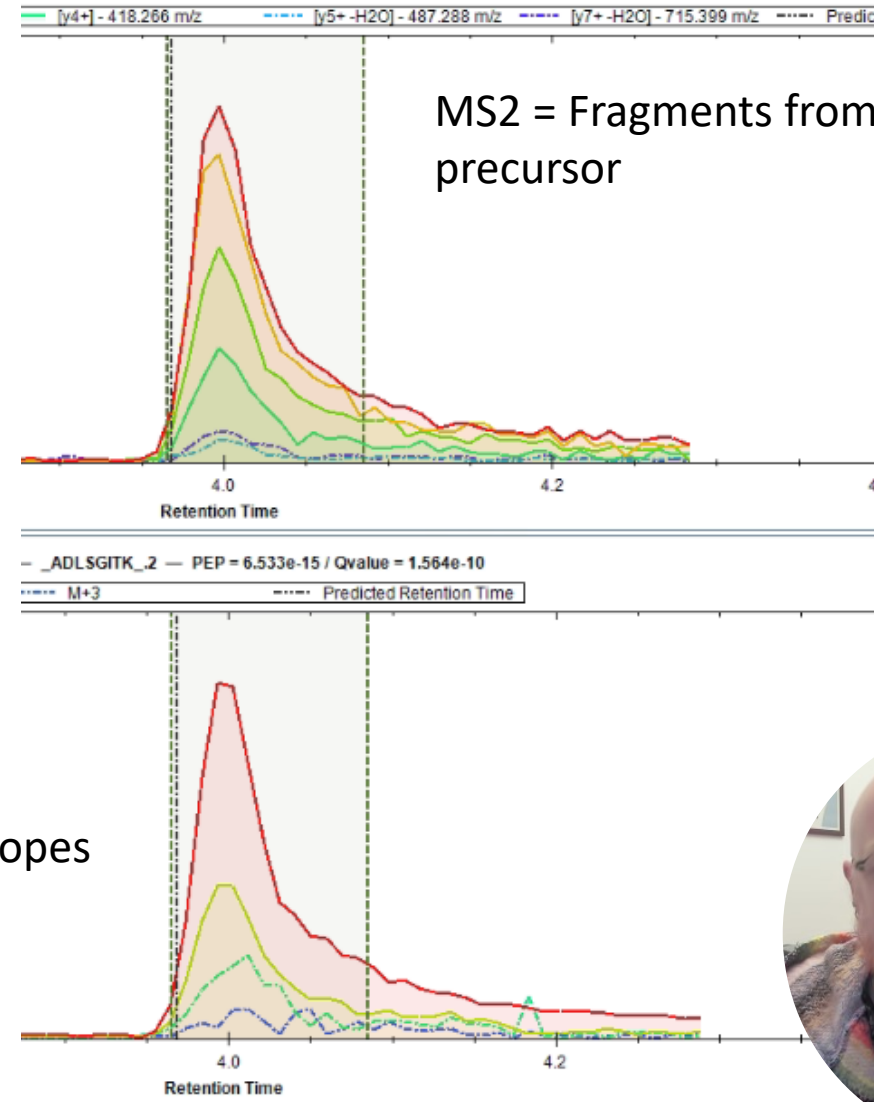
And sum them together in various clever ways



We also break apart the peptides and measure their fragments!

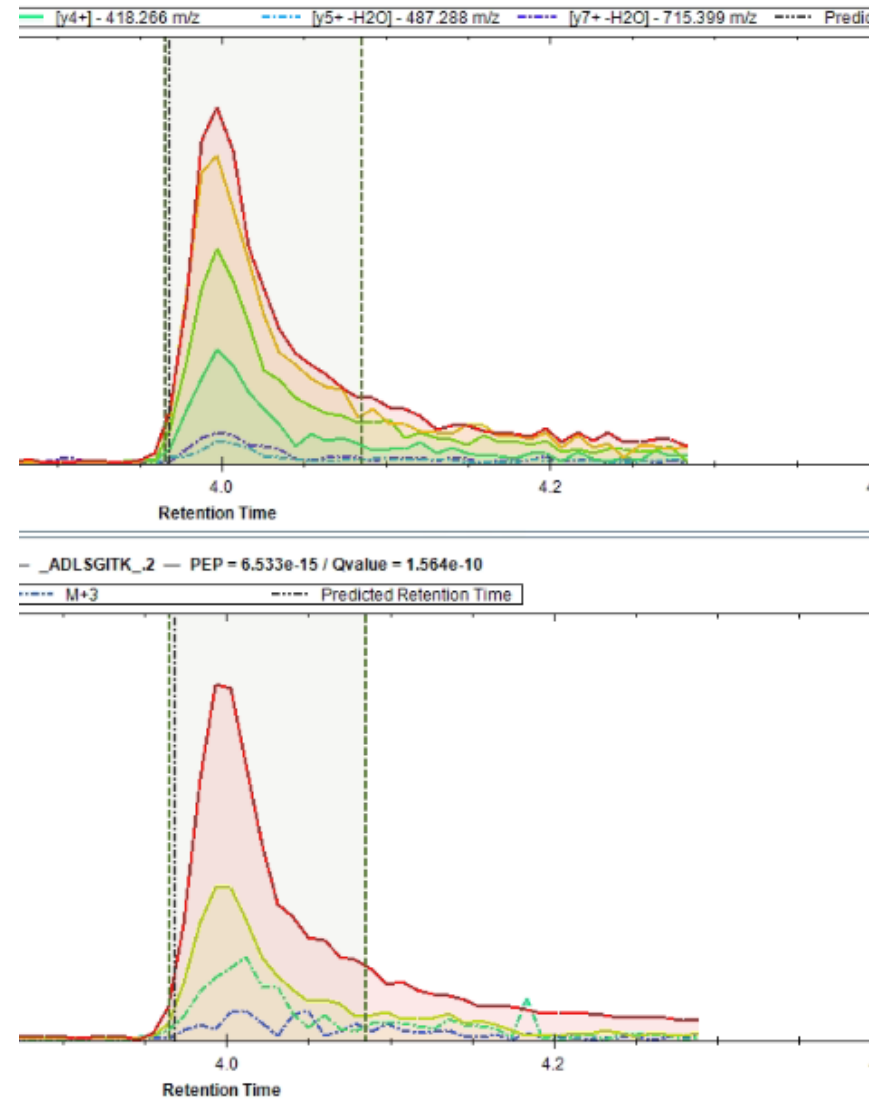


MS1 = intact precursor isotopes

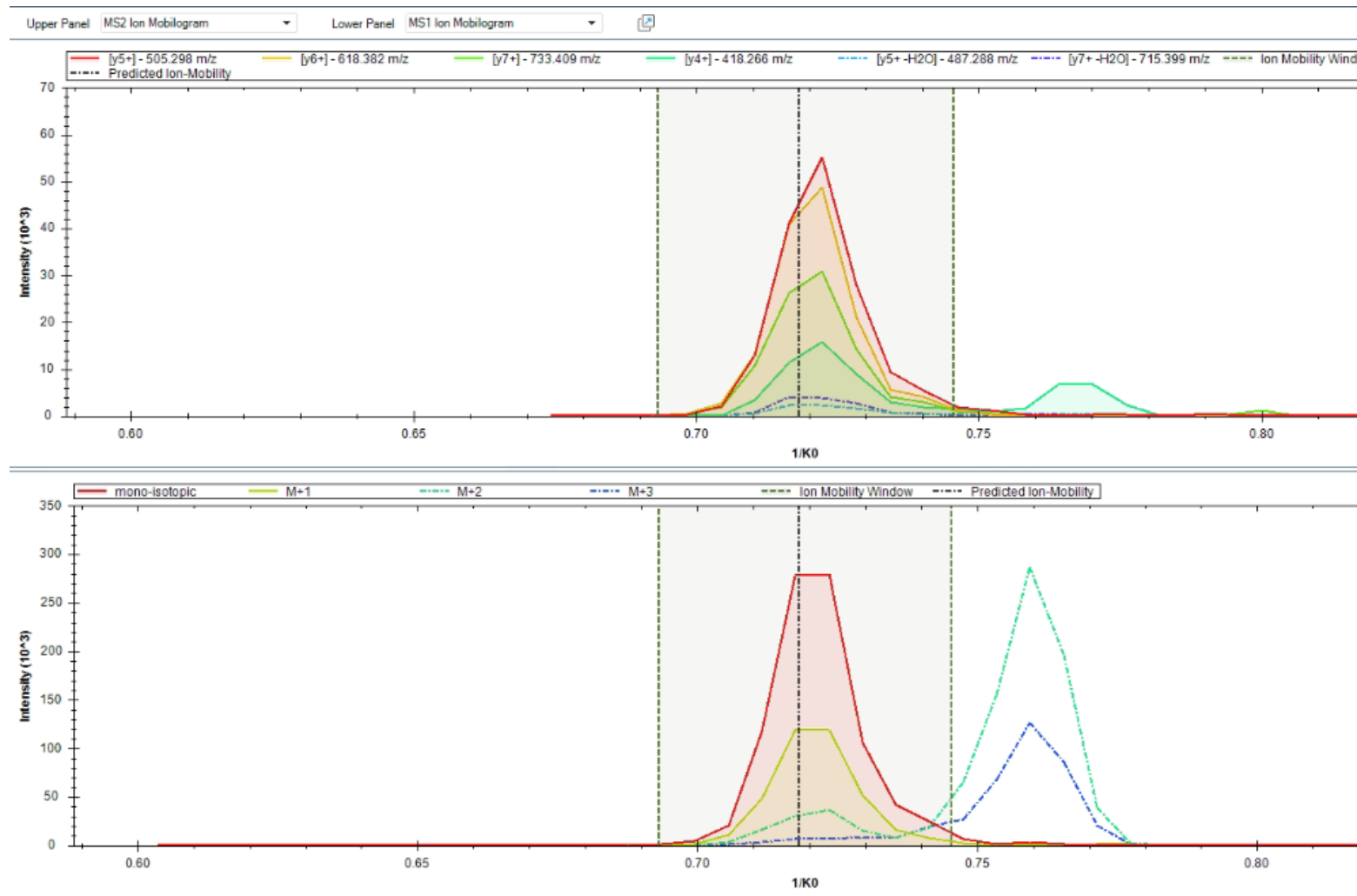


Ions also elute **together in time** if they are from the same molecule

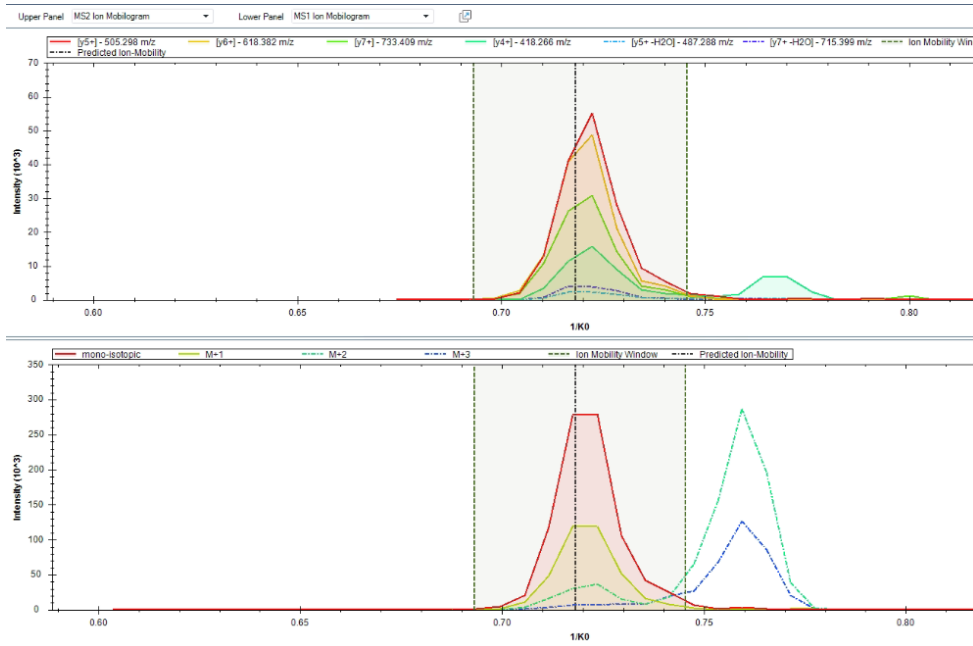
Spectronaut calls these **elution groups EG's** as the ions all elute together



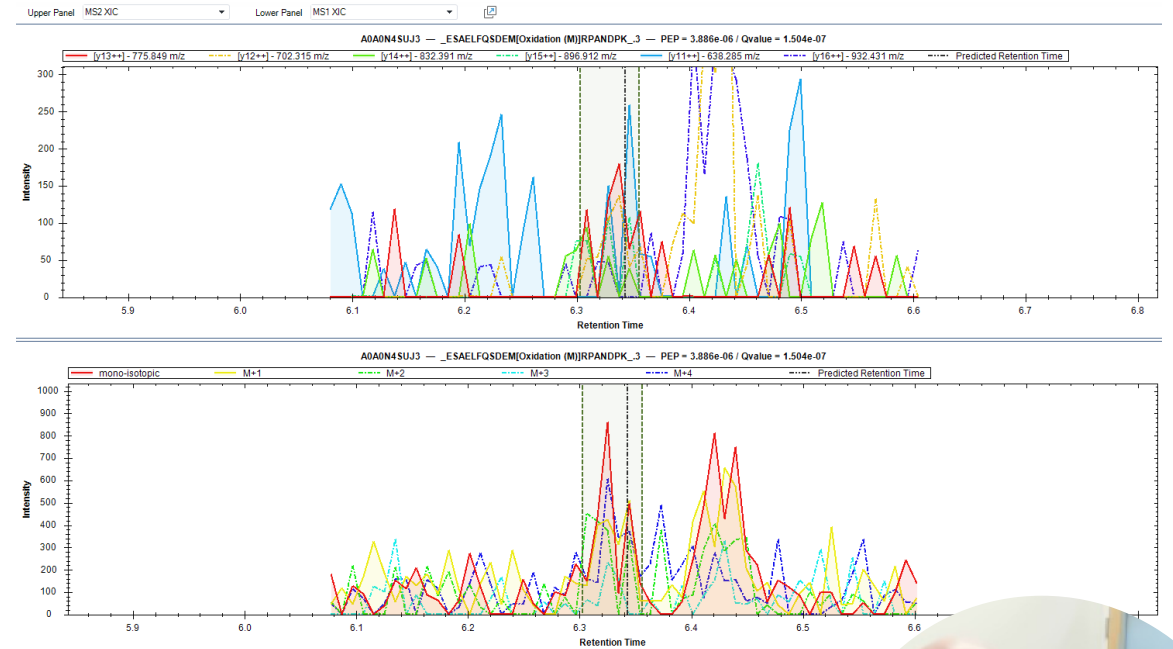
Ions also have the same ion mobility **IM** (shape) if they are from the same molecule



Many times we can detect peptide EG's we cannot quantify



This is a nice detection we can quantify



Surprisingly this mess of ions is detection. Can we quantify it? Maybe? Maybe not



Just an aside

We used to Identify peptides from their MS2 spectra (pre 2017)

Now we detect peptides using AI & statistics using MS1 ions, MS2 ions, Retention time & ion mobility and many many more metrics



So what we are typically measuring...

Area over time of Peptide Ion w/ 2 protons
= MS1 Precursor isotopes

Area over time of fragment Ions w/ 2 protons
= MS2 fragment precursors

Area over time of Peptide Ion w/ 3 protons
= MS1 Precursor isotopes

Area over time of fragment Ions w/ 3 protons
= MS2 fragment Precursor

Elution Group for +2 precursor

Elution Group for +3 precursor

Peptide (pep)

At this point we have measured
like 15 areas or intensities!!

- All the MS1 isotopes per precursor (typically 3 per precursor)
- All the Fragment ions per precursor (typically 6-7 per precursor)

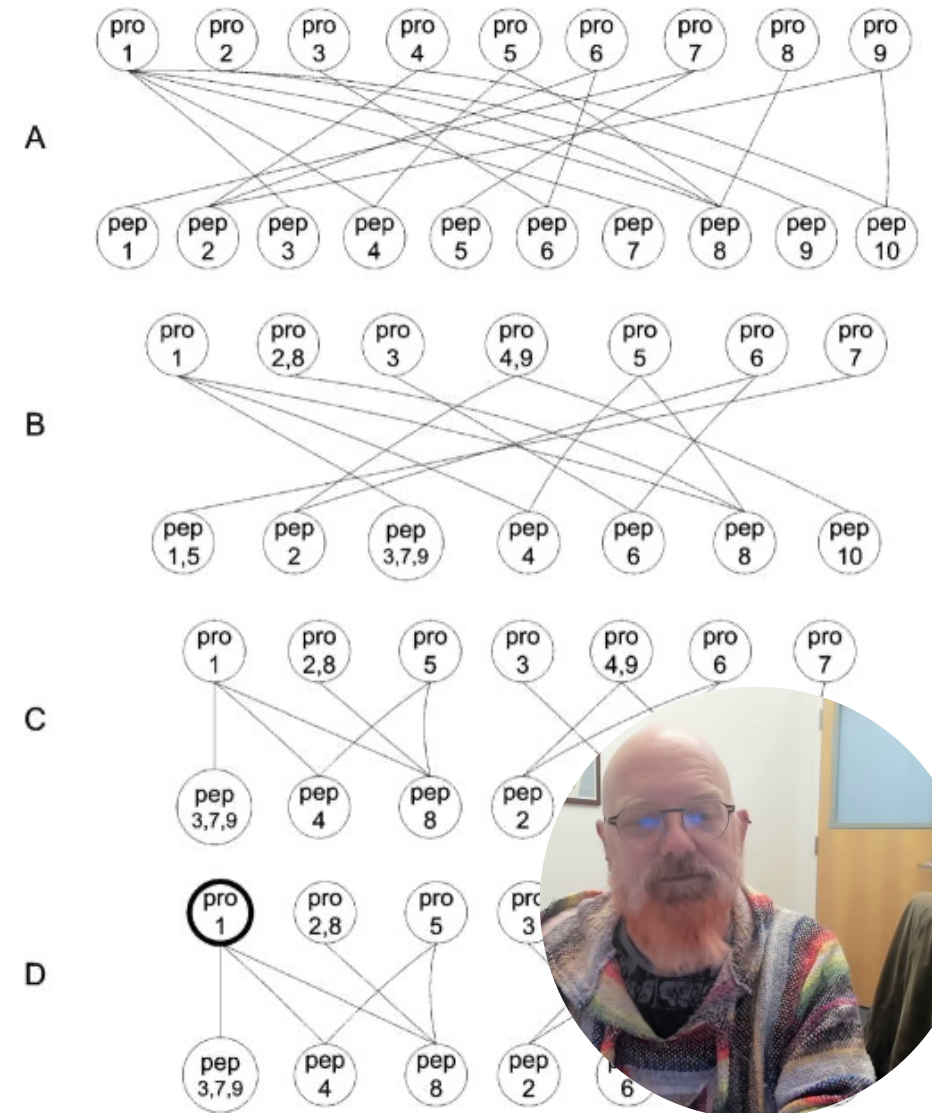


Then we do some clever things to get a protein group quantity **PG.Quantity** from the peptide quantity (**pep.quantity**) !

Spectronaut uses the IDPicker Algorithm

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2810678/>
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2810655/>

Overview of the bipartite graph model and analysis procedure for protein assembly. (A) Initialize: connect peptides to all proteins that contain them in a bipartite graph. (B) Collapse: collapse indistinguishable protein and peptide vertices together to form meta-protein and meta-peptide vertices. (C) Separate: decompose the graph into protein clusters with shared peptides. (D) Reduce: identify and highlight a minimum protein list.



Then we do some clever things to compare protein quantity across different samples

Usually most software uses
MaxLFQ

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4159666/>

Essentially it's ratios of
Precursor or Elution
group? areas across
samples

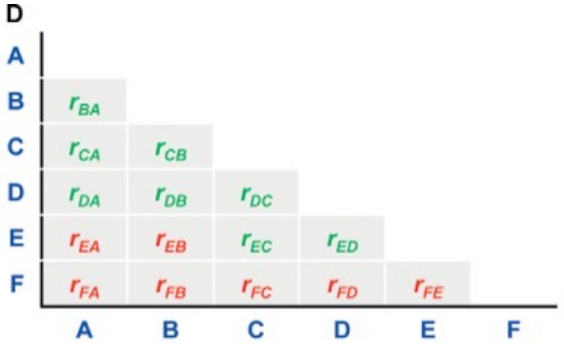
A
>P63208
MPSIKLQSSDGEIFEVDVEIAKQSVTIKTMLEDLGMDDEGDD
DPVPLPNVNAAILKKVIQWCTHHKDDPPPPEDDENKEKRTDD
IPVWDQEFLKVDDQGTLELILAAANYLDIKGLLDVTCKTVANM
IKGKTPEEIRKTFNIKNDFTTEEEEAQVRKENQWCEEK

B

Peptide species	Sequence	Charge	Mod.
P ₁	LQSSDGEIFEVDVEIAK	2	–
P ₂	LQSSDGEIFEVDVEIAK	3	–
P ₃	RTDDIPVWDQEFLK	2	–
P ₄	TVANMIK	2	–
P ₅	TVANMIK	2	Oxid.
P ₆	TPEEIRK	3	–
P ₇	NDFTEEEEAQVR	2	–

C

Sample	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇
A		+				+	
B		+	+			+	
C	+	+	+	+		+	+
D	+	+		+		+	+
E		+		+			+
F		+			+		



E

$r_{BA} = I_B / I_A$	$r_{CA} = I_C / I_A$	$r_{CB} = I_C / I_B$
$r_{DA} = I_D / I_A$	$r_{DB} = I_D / I_B$	$r_{DC} = I_D / I_C$
$r_{EC} = I_E / I_C$	$r_{ED} = I_E / I_D$	$I_F = 0$

